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Subject Code/Name: MLA0305-Reinforcement Learning

EXPERIMENT – 12

Actor-Critic Reinforcement Learning Using A2C

Aim

To implement an Actor-Critic Reinforcement Learning algorithm using Advantage Actor-Critic (A2C) and evaluate the learning performance of the agent.

Procedure

1. Create the CartPole environment.
2. Build an Actor network to select actions.
3. Build a Critic network to estimate the value of the current state.
4. Generate an episode using the Actor's policy.
5. Calculate the reward and advantage.
6. Update the Actor using the policy-gradient method.
7. Update the Critic using the value-function error.
8. Repeat the process for multiple episodes.
9. Plot the episode rewards to evaluate learning performance.

Code:

```
# ============================================================
# EXPERIMENT 12: ACTOR-CRITIC REINFORCEMENT LEARNING (A2C)
# ============================================================

import numpy as np
import tensorflow as tf
import gymnasium as gym
import matplotlib.pyplot as plt

# ------------------------------------------------------------
# Reproducibility
# ------------------------------------------------------------
np.random.seed(42)
```

```
tf.random.set_seed(42)

# ------------------------------------------------------------
# STEP 1: Create Environment
# ------------------------------------------------------------
env = gym.make("CartPole-v1")
state_size = env.observation_space.shape[0]
action_size = int(env.action_space.n)

# ------------------------------------------------------------
# STEP 2: Create Actor Network
# ------------------------------------------------------------
actor = tf.keras.Sequential([
    tf.keras.layers.Input(shape=(state_size,)),
    tf.keras.layers.Dense(64, activation="relu"),
    tf.keras.layers.Dense(64, activation="relu"),
    tf.keras.layers.Dense(action_size, activation="softmax")
])

# ------------------------------------------------------------
# STEP 3: Create Critic Network
# ------------------------------------------------------------
critic = tf.keras.Sequential([
    tf.keras.layers.Input(shape=(state_size,)),
    tf.keras.layers.Dense(64, activation="relu"),
    tf.keras.layers.Dense(64, activation="relu"),
    tf.keras.layers.Dense(1)
])

# ------------------------------------------------------------
# Optimizers
# ------------------------------------------------------------
actor_optimizer = tf.keras.optimizers.Adam(learning_rate=0.001)
critic_optimizer = tf.keras.optimizers.Adam(learning_rate=0.001)
```

```
gamma = 0.99
episodes = 100
reward_history = []

# ------------------------------------------------------------
# STEP 4: A2C Training
# ------------------------------------------------------------
for episode in range(episodes):
    state, _ = env.reset()
    total_reward = 0
    done = False

    while not done:
        state_input = np.array([state], dtype=np.float32)

        # Actor selects action
        action_prob = actor(state_input, training=False).numpy()[0]
        action = np.random.choice(action_size, p=action_prob)

        # Environment step
        next_state, reward, terminated, truncated, _ = env.step(action)
        done = terminated or truncated
        next_state_input = np.array([next_state], dtype=np.float32)

        # Critic estimates state values
        value = critic(state_input, training=False)[0, 0]
        next_value = critic(next_state_input, training=False)[0, 0]

        # TD Target and Advantage
        target = reward + (0 if done else gamma * next_value)
        advantage = target - value

        # Actor Update
        with tf.GradientTape() as tape:
            probs = actor(state_input, training=True)
```

```
            selected_prob = probs[0, action]
            log_prob = tf.math.log(selected_prob + 1e-8)
            actor_loss = -log_prob * tf.stop_gradient(advantage)

        actor_grads = tape.gradient(actor_loss, actor.trainable_variables)
        actor_optimizer.apply_gradients(zip(actor_grads, actor.trainable_variables))

        # Critic Update
        with tf.GradientTape() as tape:
            predicted_value = critic(state_input, training=True)[0, 0]
            critic_loss = tf.square(tf.stop_gradient(target) - predicted_value)

        critic_grads = tape.gradient(critic_loss, critic.trainable_variables)
        critic_optimizer.apply_gradients(zip(critic_grads, critic.trainable_variables))

        state = next_state
        total_reward += reward

    reward_history.append(float(total_reward))
    print(f"Episode {episode+1:3d} | Reward: {total_reward:.0f}")

env.close()

# ------------------------------------------------------------
# STEP 5: Results
# ------------------------------------------------------------
print("\n" + "="*50)
print("A2C RESULTS")
print("="*50)
print("Average Reward:", f"{np.mean(reward_history):.2f}")
print("Best Reward:", f"{max(reward_history):.0f}")

# ------------------------------------------------------------
# STEP 6: Visualization
# ------------------------------------------------------------
```

```
plt.figure(figsize=(10,5))
plt.plot(reward_history, label="Episode Reward")
plt.title("Actor-Critic (A2C) Learning Performance")
plt.xlabel("Episode")
plt.ylabel("Total Reward")
plt.legend()
plt.grid(True)
plt.show()

# Moving average
window = 10
if len(reward_history) >= window:
    moving_avg = np.convolve(reward_history, np.ones(window)/window, mode="valid")
    plt.figure(figsize=(10,5))
    plt.plot(moving_avg, label="10-Episode Moving Average")
    plt.title("A2C Moving Average Reward")
    plt.xlabel("Episode")
    plt.ylabel("Average Reward")
    plt.legend()
    plt.grid(True)
    plt.show()
```

OUTPUT:-

Episode   1 | Reward: 48

Episode   2 | Reward: 30

Episode   3 | Reward: 26

Episode   4 | Reward: 47

Episode   5 | Reward: 9

Episode   6 | Reward: 19

Episode   7 | Reward: 11

Episode   8 | Reward: 16

Episode   9 | Reward: 12

Episode 10 | Reward: 10

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A2C RESULTS

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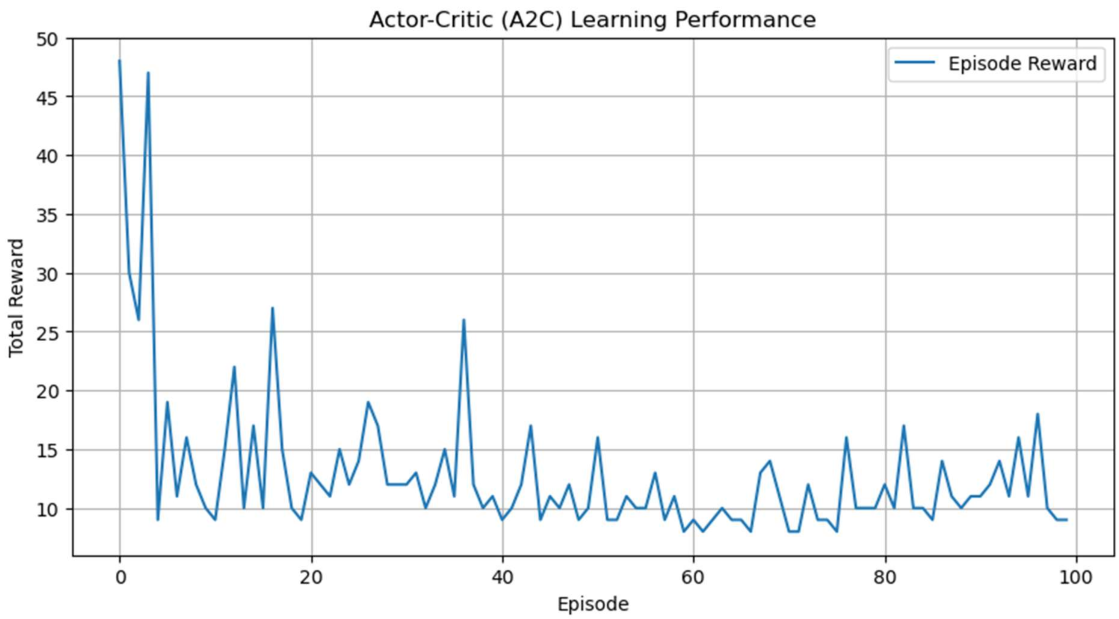
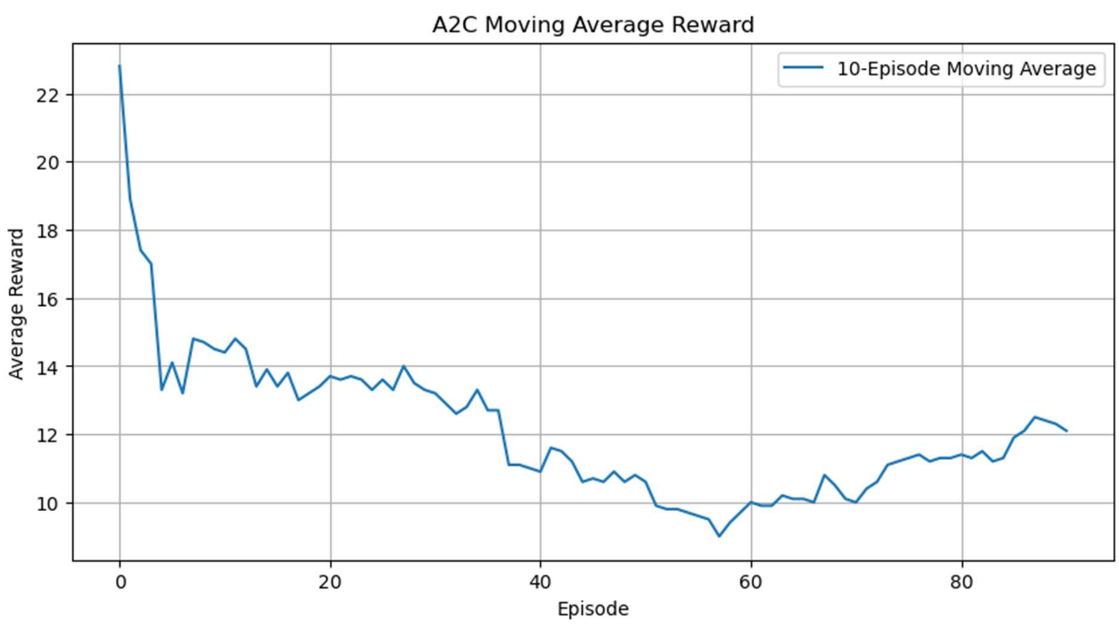
Average Reward: 12.91

Best Reward: 48





Visualization

The program generates two graphs:

1. A2C Learning Performance
   Shows the total reward obtained in each episode.
2. A2C Moving Average Reward
   Shows the average reward over 10 episodes and helps identify the overall learning trend.

Summary

The Actor-Critic (A2C) method was implemented using separate Actor and Critic neural networks. The Actor learns the action policy, while the Critic estimates the state value and provides an advantage signal for improving the policy.

Result

Thus, the Actor-Critic Reinforcement Learning algorithm using A2C was successfully implemented, and its learning performance was evaluated using reward-based visualizations.